

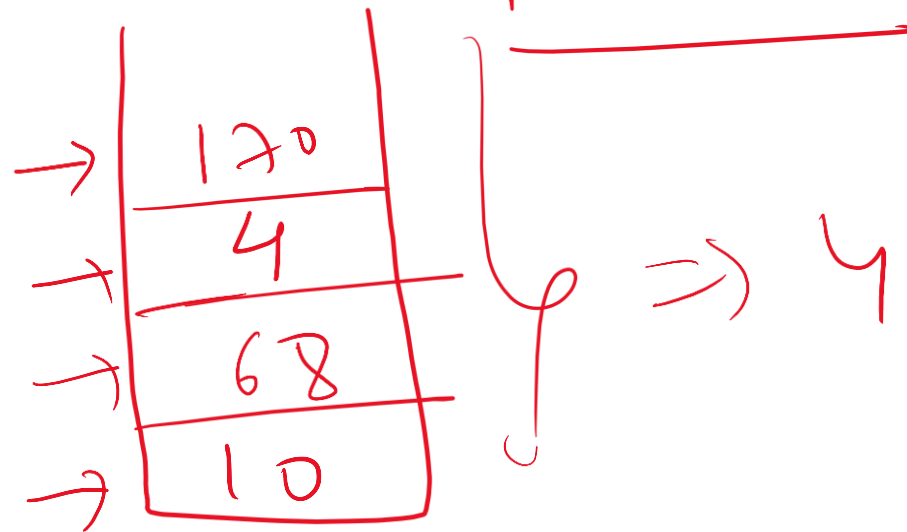
① Stack :- It is a data structure which follows the property of LIFO (Last In First out).

② $top()$ / $peek()$ $\rightarrow$ It returns top element of stack in $O(1)$.

③ $pop()$ $\rightarrow$ It deletes top element from stack in $O(1)$.

o $\text{push}() \rightarrow$ It pushed new element to the top of the stack in $O(1)$ T.C.

eg:- $\text{add}(10), \text{add}(68), \text{add}(\underline{4}), \text{add}(\underline{30}),$
 $\text{pop}(), \text{add}(170)$



① `size()` → It returns the no of elements present in stack.

① `IsEmpty()` → It returns true if the stack is empty else false.

① During an interview, whenever you feel that you need to traverse backward as compared to current index location, in that case rather than using Nested Loop (to go backward), think of Stack as a possible solution as all operation in stack takes

$O(1)$ T.C.

0	1	2	3	4	5	6
10	20	25	38	49	59	80



```
for (int i = 0; i < n; i++)
```

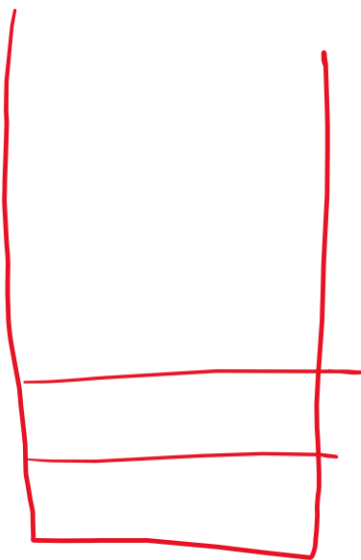
```
{
    for (int j = i - 1; j >= 0; j--)
```

```
{
```

```
    // Do some operation
```

```
}
```

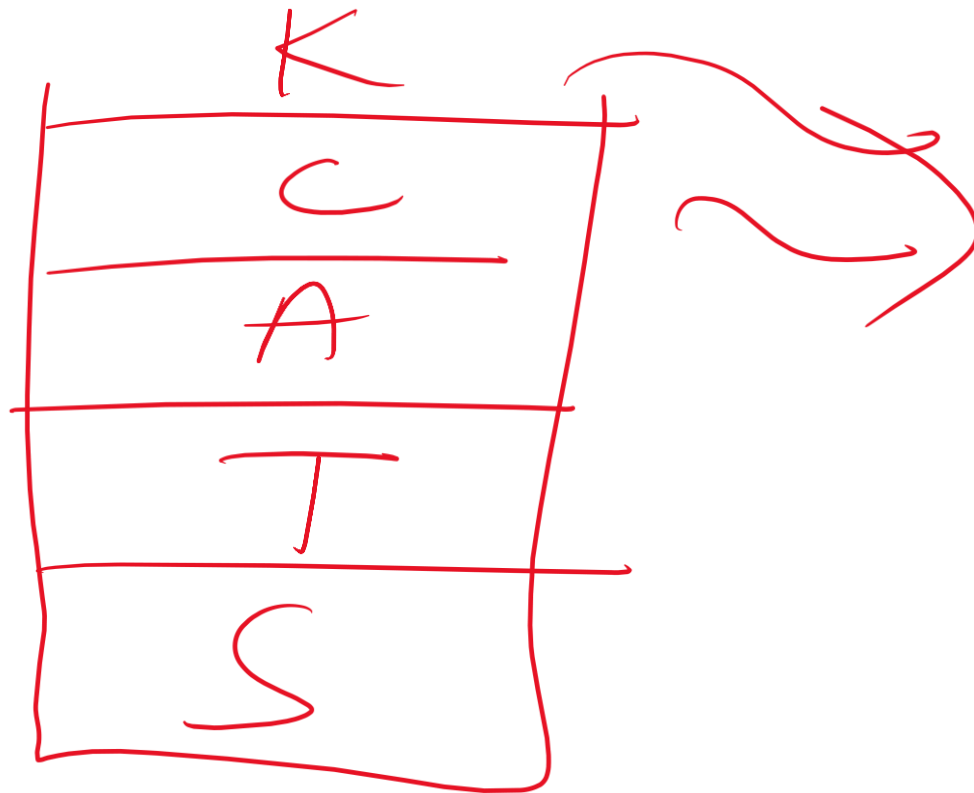
```
}
```



① Whenever you add an element
to the stack, that element
could be index, array
element or in case of string
it could be character.

① Stack can also be used to reverse the string

eg!- STACK



KCATS

Reversed word

① Problem Statement :- BACD

stack < char > st;
for (Loop)

{



while (! st.empty())

{

st.pop();

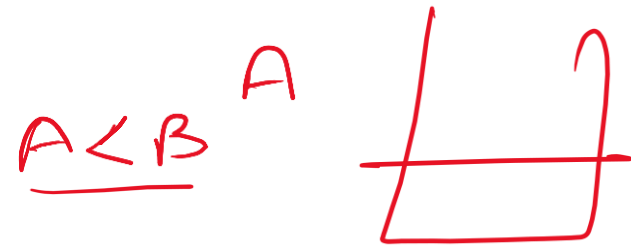
// Remove element from stack

Since it doesn't lead you
to final correct output as reqd.

}

st.push(ch);

}



ch < st.top()